Feynman Diagrams

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1 Introduction

These are notes from a lecture given by E. Witten at the IAS many many years ago. But I will add further details and facts from other papers and topics related to string theory.

2 Feynman diagrams in QFT

Consider a generic Feynman diagram which we draw abstractly as a graph consisting of edges and vertices that may or may not have half edges. To each edge (or line) we assign a propagator $\frac{1}{p^2+m^2}$ that can be written as the integral

$$\frac{1}{p^2+m^2} = \int_0^\infty dt \exp(-t(p^2+m^2)) \tag{1}$$

where t is the proper length of the line (Witten calls them proper time) or the Schwinger parameter.

For convenience the graph (which Witten denotes by Γ) can be thought of as a singular 1-manifold as it has singularities at it's vertices. Now if we assign a Schwinger parameter to each of the lines of Γ, we see that we have a Riemannian metric on it. Up to diffeomorphisms of the internal lines in Γ, the only invariant of such a metric are the (proper) lengths of the line segments that are t_i's. Although each internal line of the Feynman graph Γ has it's own momentum p_i, we do not just integrate over the momenta p_i's independently since we need to impose the momentum conservation of particle interactions. The vertices of the graph physically corresponds to the interactions of particles that are incoming/outgoing along the lines of the graph. So momentum conservation must be imposed at each vertex.
If we have n particles with momenta $p_1, ..., p_n$ that are incoming at a vertex, and m particles with momenta $p'_1, ..., p'_m$ that are outgoing from that vertex, we want a delta function

$$(2\pi)^4 \delta^4(\sum_{i=1}^{n} p_i - \sum_{j=1}^{m} p'_j) \tag{2}$$

at that vertex. Let's just use the notation $\delta^4(\sum_i p_i)$ from now on. This delta function can be obtained from the integral

$$\int d^4x \exp(\sqrt{-1}\sum_i p_i \cdot x) = (2\pi)^4 \delta^4(\sum_{i=1}^n p_i) \tag{3}$$

Now we assign a spatial coordinate to each vertex of Γ. For example, if two vertices with spatial coordinates x, y connected by a propagator $\frac{1}{p^2+m^2}$, we can write the propagator in position space as

$$G(x-y) = \int \frac{d^4p}{(2\pi)^4} \frac{e^{ip\cdot(x-y)}}{p^2+m^2} \tag{4}$$

$$= \int_0^\infty dt \frac{d^4p}{(2\pi)^4} \exp(\sqrt{-1}p \cdot x - t(p^2+m^2))$$

Thus, we now have a slightly different way to interpret a Feynman diagram: We integrate over a position parameter x for each vertex and a length parameter t for each line where each line has the factor

$$G(x-y;t) = \int \frac{d^4p}{(2\pi)^4} e^{\sqrt{-1}p\cdot(x-y)-t(p^2+m^2)} \tag{5}$$

such that

$$G(x-y) = \int_0^\infty dt G(x-y;t)$$

Recall what Feynman taught us about interpreting the function (5). We think about a non relativistic point particle with Hamiltonian

$$H = p^2 + m^2$$

The action for such a particle is

$$I = \int [\dot{x}^2 + m^2] dt$$

which is just the action for a non relativistic point particle with $2m = 1$ and a constant m^2 added to the Lagrangian density.
Then according to Feynman, the position space propagator $G(x-y;t)$ can be obtained as an integral (whose integration measure doesn't have a rigorous definition) in the space of all paths $X(t')$ (which being a function space is an infinite dimensional space) such that $X(0) = y$, $X(t) = x$ (i.e. all paths by which the particle travels from y to x in time t):

$$G(x-y;t) = \int DX(t') \exp(-\int_0^t dt' [\sum_i \dot{X}^2 + m^2]) \tag{6}$$

where the index i runs from 1 to the number space time dimension. This is the basic Feynman path integral of non-relativistic quantum mechanics.

Summarizing the above, when we evaluate a Feynman diagram we have to integrate over all possible Riemannian metrics on the graph which is equivalent to integrating over all the (proper) length parameters t_i's. Then we also have to integrate over all possible locations of the vertices x's in space time and also over all possible maps of the lines connecting these vertices into space time. This means, all we do is that we integrate over all possible maps of the Feynman graph Γ into space time.

So to evaluate the amplitude associated to a Feynman graph Γ we need to

1. integrate over all possible (piece-wise) metrics on Γ on the edges of Γ modulo diffeomorphisms acting the edges.

2. integrate over all possible maps from Γ to space time.

This is essentially a version of 1-dimensional General Relativity (GR) where the fields are

1. a metric on Γ

2. a map from Γ to space time

Denoting h the 1×1 metric tensor of Γ and g the $d \times d$ metric tensor of space time, the relevant Einstein-Hilbert action for this 1-dimensional GR is

$$I = \int_\Gamma ds \sqrt{\det(h)}(h^{-1} \sum_{ij} g_{ij} \frac{dX^i}{ds} \frac{dX^j}{ds} + m^2) \tag{7}$$

Remarks:

1. There is no purely Einstein-Hilbert action $\int_\Gamma ds \sqrt{\det(h)} R$ as there is no Ricci curvature in 1-dimension.

2. the integral over metrics can be reduced to an integral over the Schwinger parameters t_i's via using the gauge $h = 1$

3. the target manifold (i.e. the space-time manifold) does not have to be a flat manifold.

Coming back to the propagators, we note an important point that is, the integral over each of the Schwinger parameters t's has two ends. There is the limiting case $t \to \infty$ that generates the pole of the propagator:

$$\int_\Lambda^\infty dt \exp(-t(p^2 + m^2)) \sim \frac{1}{p^2 + m^2} \tag{8}$$

we don't want to do without this region since the physical interpretation of QFT depends crucially on the pole of the propagator.

Secondly, the other end of t integral is responsible for the fact that propagator is singular at short distances:

$$\int_0^\Lambda dt \int \frac{d^d p}{(2\pi)^d} e^{\sqrt{-1}p\cdot(x-y)-t(p^2+m^2)} \sim \frac{1}{|x-y|^{d-2}} \tag{9}$$

which shows that singular short distance behavior of the propagator comes completely from the small t part of the integral. This is where UV divergences come from. When all the proper time parameters in a loop go to zero and all the vertices in the loop map to (almost) the same point in space-time, we potentially run into UV divergences.
In qft there is no way to get rid of the small t region because if we put a lower cuttoff on t, we would spoil space-time locality. This is the reason that QFTs are at risk of UV divergences.

3 Strings

Moving on to string theory, we elevate by one the dimension of our source manifolds i.e. from graphs we now move to considering surfaces of a particular type called Riemann surfaces, metrics on these surfaces and maps from these to space-time; because whereas a point particle sweeps a one dimensional trajectory, a string sweeps out a 2d surface that we call the worldsheet of the string. Plus, instead of 1-dimensional GR, our study has now been promoted to 2d GR. Unlike the previous, this time the manifold is smooth. There is no need to assume singularities and a lot of the machinery of QFT will not have analogs. Feynman graphs were divides into different lines that can represent different particles with different masses and spins. But in the string case, any part of the string worldsheet is equivalent to any other so "there is only one string". Whatever particles there are, are going to be represented by different states of vibration of one basic string. Also, we don't have vertices in the worldsheet so we do not have the freedom to tell the string how to interact.
Unlike 1-manifolds, 2-manifolds can have non zero Ricci curvature. The space of all the possible metrics on a smooth 2-manifold modulo coordinat transformations is an infinite dimensional space (another manifold).
Being a symmetric matrix, the metric tensor

$$h = \begin{bmatrix} h_{11} & h_{12} \\ h_{12} & h_{22} \end{bmatrix}$$

has 3 independent component, while a diffeomorphism generator

$$x^i \to x^i + \epsilon^i(x) \tag{10}$$

where $i = 1, 2$ only depends on 2 functions. This is not enough to gauge fix the three components of h as one field is left over. One way to proceed is to accept the fact that we will have to do a path integral over this field, and try to

make sense of it. Witten suggests that there is an alternative route, which tells us to impose an extra symmetry that eliminates 1 component of h. That extra symmetry on the metric is

$$h_{ij} \simeq h_{ij} e^{2\sigma} \tag{11}$$

known as the conformal or Weyl symmetry, for any smooth function σ on the worldsheet. This symmetry preserves the angles between any two curves drawn on the surface, while the lengths of the curves might change after this symmetry transformation. The reason for considering Riemann surfaces, that are surfaces on which we can define holomoprhic functions, is that for such surfaces the quotient space of metrics modulo conformal transformations is a finite dimensional space. There are many ways of defining holomorphic functions and hence do complex analysis on *nice* surfaces, each of such ways are called a complex structure. There is a whole continuous space of all the possible complex structures that can be equipped on a genus g smooth surface called the moduli space of complex structures which has complex dimension $3g - 3$ for $g > 1$. Denote the complex structure parameters (i.e. coordinates on the moduli space) by τ_i (whihc are complex numbers) of the Riemann surface which is the worldsheet of a string. These τ_i's are analogous to the proper time or Schwinger parameters t_i's as in the case of Feynman graphs. So when we follow our previous procedure as in the case of Feynman grpahs, instead of having an infinite dimensional integral over the space of Riemannian metrics, we are having a finite dimensional integral over the space of all possible complex structures on the Riemann surface.

Just as we did in 1d GR, we are going to add "matter" to our 2d GR. Since we are using the conformal invariance to reduce the infinite dimensional integral to a finite dimensional one, the "matter" part of the action has to be conformally invariant. For instance, the usual action for massless scalar fields

$$I = \int_\Sigma d^2\sigma \sqrt{\det g} (g^{\alpha\beta} \partial_\alpha X^I \partial_\beta X^J G_{IJ}(X)) \tag{12}$$

where the fields are maps $X : \Sigma \to M$, G_{IJ} being the space-time metric.

There is still (almost) no purely gravitational action in 2d since the Einstein-Hilbert action in this case

$$\int_\Sigma d^2\sigma \sqrt{\det h} R \tag{13}$$

is the Gauss-Bonnet integral which equals to $2 - 2g$ where g being the genus of the surface Σ. Thus just as in ordinary QFT, we basically have to only consider the action for the matter fields.
To develop the theory, we are supposed to do the following:

1. for a fixed Σ, we perform a Feynman path integral over the fields $X = (X^1, ..., X^D)$
2. then integrate over the moduli τ_i's

3. sum over all topological choices for Σ

the last step is analogous to summing up over all topologically distinct Feynman graphs in ordinary QFT. One of the many advantages of the string theoretic path integrals is that the UV divergences that we saw before in QFTs don't show up here. To understand that let us consider the 1-loop contribution to the vacuum energy. An example of a divergent 1-loop contribution to the vacuum energy comes from a scalr field of mass m. It's path integral gives (in ordinary QFT)

$$\frac{1}{\sqrt{\det(-\nabla^2+m^2)}} = \exp(-\frac{1}{2}\operatorname{Tr}\log(-\nabla^2+m^2)) \tag{14}$$

meaning that the 1-loop contribution to the effective action is

$$I^* = \frac{1}{2}\operatorname{Tr}\log(-\nabla^2+m^2) = \frac{1}{2}\int_0^\infty \frac{dt}{t}\exp(-tH) \tag{15}$$

where $H = p^2 + m^2 = -\nabla^2 + m^2$. Here the length parameter t is the circumference of a loop. Clearly, this integral is divergent at $t = 0$. This is another example of UV divergences.

So what happens in string theory, when we try to do something similar? In string theory, the circle is replaced by a torus, which can be (topologically) constructed from a rectangle with a flat metric. We label the horizontal sides by s and the vertical ones by s'. We could view the rectangle as describing an open string of circumference s propagating through a proper time s' or a string of circumference s' propagating a proper time s. Either picture is correct. In any event, in string theory, because of conformal invariance, only the ratio s/s' or s'/s is meaningful, as conformal transformation only deforms the lengths, not the angles between any two curves.
The string theory formula will reduce approximately to field theory if, say $s >> s'$. Then we can think in terms of a string of circumference s' propagating for a proper time s. Because of conformal invariance we can set $s' = 1$ and identify s with the integral that in field theory is an integral over the proper time t is in string theory replaced by an integral over the ratio s/s'.
The difference between field theory and string theory comes in because in string theory we have the symmetry

$$s \longleftrightarrow s'$$

Because of this symmetry, we are free to restrict the integration region to $s \geq s'$ or equivalently we can restrict ourselves to $t \geq 1$ where $t = s/s'$.
Thus, in field theory when we evaluate a 1-loop diagram, we integrate the Schwinger parameter t ranging from 0 to ∞ and we typically get a UV divergence at $t \to 0$ i.e. as the loop gets smaller and smaller, the amplitude gets bigger and bigger. Depending on the theory, we may find IR divergences for $t \to \infty$. In string theory, we integrate over the *reduced Schwinger* parameter t ranging from 1 to ∞. Since it begins at 1, there is no UV divergence. Depending

on the theory, s there still may be IR divergences for $t \to \infty$.

Above, what Witten explained is in a slightly naive way, but the point holds to be true in general. The moduli space of Riemann surfaces has a region "at infinity" that corresponds to the infrared region $t \to \infty$ in field theory. But it has no region corresponding to the ultraviolet region $t \to 0$ in field theory. So string theory is free of the UV divergences of field theory.
Another non trivial fact that Witten mentions is the following: The IR behavior of string theory matches the IR behavior of a field theory with appropriate light particles and interactions.

Strings and Gravity

So why does string theory describe gravity in the target space time, which we denote here by M? It happens *automatically*. Witten then shows a couple of steps.
First of all, what are the fields in target space theory? In the familiar case, they

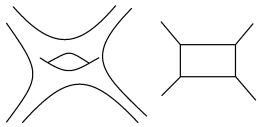


Figure 1: The string diagram (left) and the ordinary QFT diagram(right)

are the external lines in a Feynman diagram. So in the present case, the fields in the target space are the external strings (The legs of the string diagram in Figure 1). In other words, the fields are the vibrational states of a string.

If we make an infinitesimal variation of the space time metric G_{IJ} on M i.e. $G \to G + \delta G$, the consequent variation in the action is

$$\delta I = \int_\Sigma d^2\sigma \sqrt{\det h}(h^{\alpha\beta}\partial_\alpha X^I \partial_\beta X^J \delta G_{IJ}(X)) \tag{16}$$

This causes an operator insertion on the Riemann surface somewhere (say a point, that we marked cross in Figure 2). For the case that the change in the action comes from a change in the metric in space time, the operator is

$$\mathcal{O} = h^{\alpha\beta}\partial_\alpha X^I \partial_\beta X^J \delta G_{IJ}(X) \tag{17}$$



Figure 2: A generic worldsheet with a marked point (the cross)

Because of conformal invariance, the point where the operator has been inserted, can be projected to infinity (the external tube in Figure 3) So it corresponds to one of the external fields and if we shift the expectation value of that field, this amounts to shifting the metric G_{IJ} in spacetime. Thus the dynamics of gravity in spacetime is part of what string theory describes. A shift in the metric G_{IJ} is a shift in the expectation value of one of the string fields. In other words, one of the string fields is the spacetime metric.



Figure 3: An external tube on the marked point (the cross)

Going down this road (and incorporating spacetime supersymmetry to avoid some IR problems), one arrives at a systematic way to calculate quantum processes involving gravitons, free of the UV divergences that one gets if one tries to quantize GR directly. The UV divergences are absent because 2d conformal invariance completely eliminates the UV region from the Feynman diagrams.